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General Information

Title

Open Data Index for Schools (ODIS)

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Abstract/Description

This dataset is used to calculate the Open Data Index for Schools (ODIS), which offers educators and researchers a resource for contextualizing student-learning outcomes through a set of school-neighborhood indicators as well as a composite index. The underlying repository is designed to consolidate data on communities surrounding the approximately 23,000 public high schools (including magnet schools, charter schools, and traditional public schools) across the United States and make them easier for users to access and use. The indicators and composite index represent the level of "stress" that school communities experience, measured across five domains Economics, Education, Health, Housing, and Crime. Each domain comprises several indicators, each with an associated measure. The technical report provides detailed information about each variable in the dataset, along with methodological information on how to standardize, aggregate, and weight diverse data sources from census tracts, cities, and counties to create a composite index.

Point of Contact for Dataset

- For questions about the data contact Angela Hawken via hawken@jhu.edu or Jonathan Kulick via jonathan.kulick@jhu.edu or Raj Choudhary via raj@jhu.edu. (School of Government and Policy, Johns Hopkins University)
- For questions about access to the data contact Johns Hopkins University Data Services via dataservices@jhu.edu. (Johns Hopkins University Data Services)

Methodological Information

Description of methods used for collection/generation and processing of data

ODIS provides data at the School Attendance Boundary (SAB) level, using SAB shapefiles available from the National Center for Education Statistics. None of the input data are available at the SAB level, so input data are synthesized to the SAB level. Input data are available at, variously, the census-tract, large city, or county level. An SAB typically includes all or part of multiple census tracts and lies entirely within a large city or county. Census-tract-level data are synthesized to the SAB level data by an area-weighted average of the values in the census tracts intersecting the SAB; city- and county-level data are attributed to all SABs within the jurisdiction. SAB-level input data are then scaled (variously, truncated, re-centered, and expanded) so that all scaled component values are on a 0–100 scale. ODIS scores in five domains are assembled as weighted averages of the scaled component values. See the Technical Report for detail on each domain and component.

Geographic location of data collection

USA

Access Information

Other publicly accessible locations of the data

<https://odis.naf.org/>

Previous version of the data

Hawken, A., Minar, N. J., Londoño López, C., Arango-Franco, S., Choudhary, R., & Kulick, J. (2025). Open Data Index for Schools (ODIS) [Data set]. New York University. <https://doi.org/10.58153/t94md-edc80>

Data derivation sources

- National Center for Education Statistics (NCES)
 - Schools Points - https://nces.ed.gov/arcgis/rest/services/CCD/CCD_Data/MapServer/5/
 - School Attendance Boundary - <https://nces.ed.gov/programs/edge/sabs>
- US Census Bureau
 - Tracts
 - <https://www2.census.gov/geo/tiger/TIGER2023/>
 - Relationships between ZIP codes and Tracts
 - https://www2.census.gov/geo/docs/maps-data/data/rel/zcta_tract_rel_10.txt
 - <https://www2.census.gov/geo/tiger/TIGER2023/ZCTA520/>
- U.S. Bureau of Labor Statistics
 - Labor Force Data - <https://www.bls.gov/lau/tables.htm#cntyaa>
- Vera Institute of Justice
 - Incarceration Trends Dataset - <https://github.com/vera-institute/incarceration-trends/blob/master/README.md>
- County Health Rankings
 - <https://www.countyhealthrankings.org/health-data/methodology-and-sources/data-documentation>
- City Health Dashboard
 - <https://www.cityhealthdashboard.com/>
- American Community Survey
 - American Community Survey 5-Year Data (2009-2023)
- <https://www.census.gov/data/developers/data-sets/acs-5year.html>

File Level Information

Relationship between files /folder

`index_scores_v3_2026.csv` - The data provide information on neighborhoods surrounding public high schools in the United States (synthesized to the level of the School Attendance Boundary). Information is provided in five domains: economic, housing, education, health, and crime. Race/ethnicity does not enter into the index calculation, but is included in this version for context.

`ODIS Technical Report March 2026.pdf` – The technical report provides detailed information about each variable in the dataset, along with methodological information on how to standardize, aggregate, and weight diverse data sources from census tracts, cities, and counties to create a composite index.

Tabular data descriptions

Number of variables: 56

Number of cases/rows: 23,599

Variable List:

Column name	Description
NCESSCH	The 12-digit school identification number assigned by NCES
Name	The name of the school
School District	The school district of the school
State	The state of the school
FIPS County Code	The 5-digit Federal Information Processing System (FIPS) county code of the school's county
County	The county of the school
City	The city of the school
Zip Code	The zip code of the school
SAB Available	Indicates whether School Attendance Boundary is available
Gini Index	Measure of income inequality
Economic	The weighted average of Unemployment, Poverty, Access to broadband internet, and Single-parent households
Education	The weighted average of Less than HS, 2-year college or higher, and Linguistic isolation

Health	The weighted average of Access to healthcare, Infant mortality rate, SNAP recipients, Low birth weight, and Lead exposure risk
Housing	The weighted average of Housing vacancy rate, Housing affordability, and Park access
Crime	The weighted average of Violent crime rate and Incarceration rate
Composite Score	The weighted average of Economic, Education, Health, Housing, and Crime
Economic Percentile Rank	The percentile value of Economic
Education Percentile Rank	The percentile value of Education
Health Percentile Rank	The percentile value of Health
Housing Percentile Rank	The percentile value of Housing
Crime Percentile Rank	The percentile value of Crime
Composite Score Percentile Rank	The percentile value of Composite Score
Economic Median	The median value of Economic
Education Median	The median value of Education
Health Median	The median value of Health
Housing Median	The median value of Housing
Crime Median	The median value of Crime
Composite Score Median	The median value of Composite Score
Unemployment	The percentage of unemployed adults (16+)
Poverty	The percentage of children (0–17) in poverty
Access to broadband internet	The percentage of households with subscription to any broadband service
Single-parent households	The percentage of children (0–17) living with a single parent
Linguistic isolation	The percentage of limited English-speaking households
Access to healthcare	The percentage of children (6–18) with health-insurance coverage

Infant mortality rate	The number of all infant deaths (within 1 year) per 1,000 live births
SNAP recipients	The percentage of households with children under 18 receiving SNAP
Low birth weight	The percentage of live births with birth weight < 2.5 kg
Lead exposure risk	Lead exposure risk index
Housing vacancy rate	The percentage of housing units vacant
Housing affordability	The percentage of households that spend 30% or more of their income on housing
Park access	The percentage of population living within a 10-minute walk of green space
Violent crime rate	The number of reported violent-crime offenses per 100,000 population
Incarceration rate	The jail incarceration rate per 100,000 residents (15–64)
Less than HS	The percentage of adults (25+) having a high-school diploma or equivalent
2-year college or higher	The percentage of adults (25+) having a 2-year college degree or higher
2-year college	The percentage of adults (25+) having a 2-year college degree
4-year college	The percentage of adults (25+) having a 4-year college degree or higher
Graduate or professional degree	The percentage of adults (25+) having a graduate or professional degree
White alone	The percentage of White alone
Black or African American alone	The percentage of Black or African American alone
American Indian and Alaska Native alone	The percentage of American Indian and Alaska Native alone
Asian alone	The percentage of Asian alone
Native Hawaiian and Other Pacific Islander alone	The percentage of Native Hawaiian and Other Pacific Islander alone
Some other race alone	The percentage of some other race alone

Two or more races	The percentage of two or more races
Hispanic or Latino	The percentage of Hispanic or Latino

Missing data codes:

Missing values in the input sources are presented as N/A. Values that cannot be calculated due to missingness are represented as Null.